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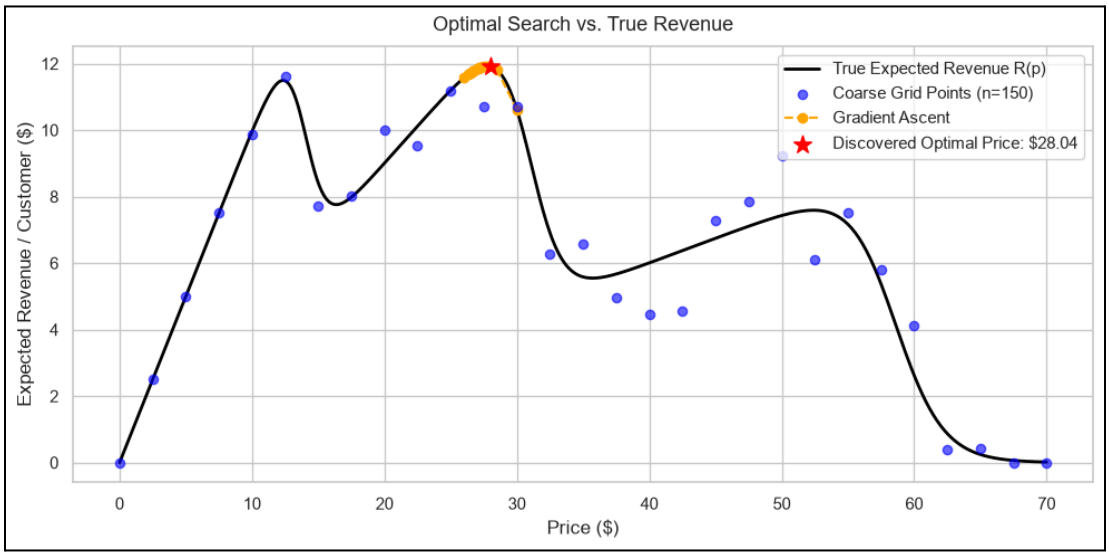
October 30th, 2026

Search Algorithm for Optimal Price in Revenue Management

In real-world market events, sellers seek to maximize expected revenue without having specific knowledge of statistical measurements, such as the underlying consumer purchase probability curve or demand function. Because customer responses are observed as stochastic Bernoulli trials (0 = no purchase, 1 = purchase), individual price tests can result in noisy revenue observations. This experiment aims to design a search algorithm that finds the global revenue-maximizing price within the bounded price interval of $0 to $70. Crucially, the algorithm must consider how to navigate the bias-variance tradeoff, gathering enough customer samples across candidate prices to mitigate statistical uncertainty without wasting customer budget at nonoptimal price points, thereby minimizing opportunity cost and cumulative regret.

To balance global and local precision, my search algorithm utilizes a hybrid search approach, combining a coarse grid search, a stochastic gradient ascent, and a fine grid search. The algorithm initiates with a coarse grid search across prices from $0 to $70 at step intervals of p = 2.50, sampling a small baseline allocation (n = 150 customers per arm) to identify optimal neighborhoods while avoiding low-performing regions. The top three candidate price regions are then reevaluated with expanded allocations (n = 500 customers) to identify the most promising starting neighborhood. From the best candidate price, the algorithm runs 15 iterations of stochastic gradient ascent using local finite differences between nearby arms (p-h and p+h, with h = 0.5). To ensure consistent convergence near our peak revenue, the learning rate decays according to $l_t = l_2\sqrt{t + 1}$, with price moves limited to $\pm$1.50 units. Finally, the algorithm conducts a fine grid search (step size p = .1) within $\pm$$1 of the gradient peak to pinpoint the optimal price.

Figure 1: Optimization Search Path vs. True Revenue Surface



Uncertainty regarding the local revenue's slope is managed through dynamic sample sizing and parametric statistical inference grounded in the central limit theorem. At each gradient ascent step 't', the customer sample allocations expand incrementally (n = 300 + 20t) to increase statistical power as price updates make finer adjustments. For each local experiment, sample means and variances are evaluated across upper and lower arms to calculate the local gradient estimate $\widehat{g_t} = \frac{\hat{r}(p+h) - \hat{r}(p-h)}{2h}$ and its standard error SE($\widehat{g_t}$). Using the standard Normal approximation, two-sided 90% Z-test confidence intervals are constructed as $\hat{g}_t \pm 1.645 \times SE(\hat{g}_t)$. The algorithm monitors these pointwise intervals against the null hypothesis $H_0: g = 0$. Early search steps exhibit statistically significant gradients whose confidence intervals strictly exclude zero, suggesting more aggressive price adjustments. As the algorithm approaches the revenue peak, the local gradients diminish, and the confidence intervals expand to encompass zero, confirming statistical convergence near the true optimum (R'(p) = 0) where power is naturally low (Figure 3).

Figure 2: KDE of Discovered Optimal Prices and ECDF of Maximum Profit Captured

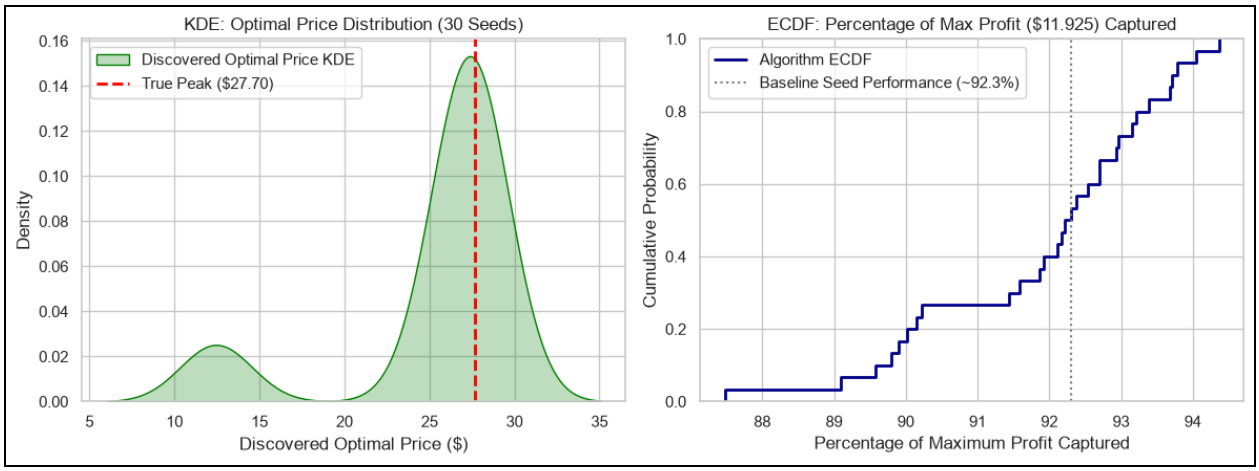
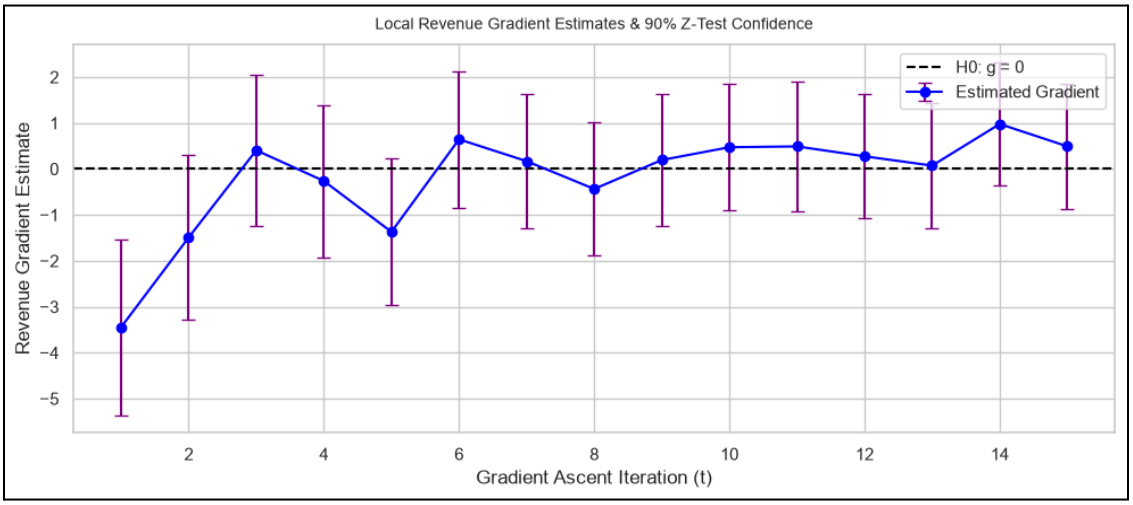


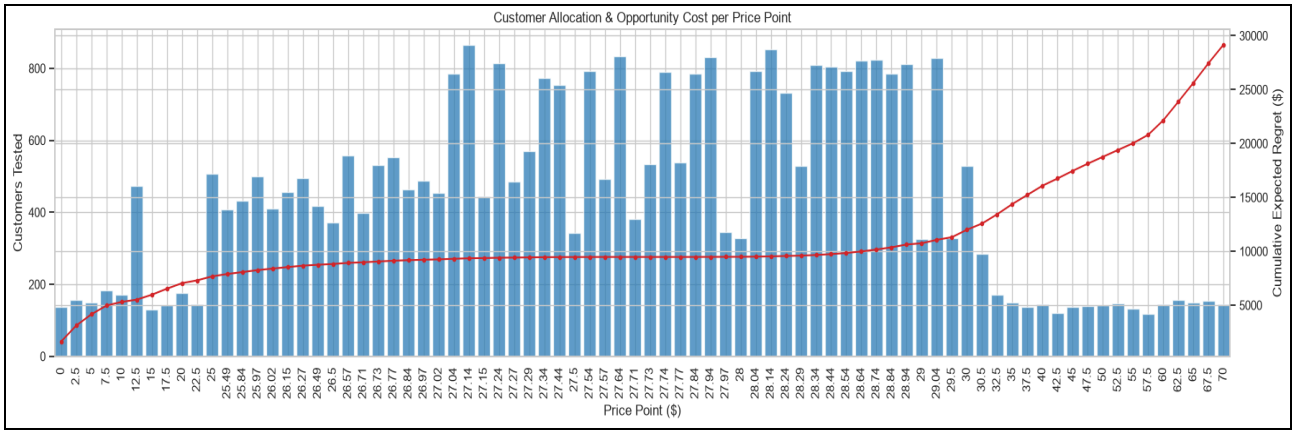
Figure 3: Local Revenue Gradient Estimates under 90% Z-Test Confidence



A primary challenge encountered during the algorithmic design process was a lack of structure when evaluating gradient convergence performance given statistical uncertainty. I had initially calculated point estimates for local revenue gradients and standard errors across 15 iterations of gradient ascent, but failed to record or evaluate the two-sided 90% Z-Test confidence interval along the active search path. To resolve this, my current algorithm modified the 'path_history' to explicitly log the confidence interval lower bound ('ci_low') and upper bound ('ci_high') at each step. I also integrated a diagnostic error-bar plot (as seen in Figure 3) displaying local gradient estimates with 90% confidence bounds against $H_0: g = 0$. This visualization confirmed that early gradient steps executed statistically significant upward price adjustments, while later iterations stabilized near the peak as the confidence intervals adjusted.

A second major challenge was evaluating the estimator stability across random streams and quantifying the trade-off between customer sample allocation and cumulative regret. My initially constructed algorithm evaluated performance solely on a single static random seed, leaving the strategy unverified against seed-dependent sampling variation and offering no proof of how customer testing budgets impacted opportunity cost. My current algorithm resolves this flaw by returning the grid ('grid_df') alongside the tracker object. Using Kernel Density Estimation (KDE) and Empirical Cumulative Distribution Function (ECDF) plots (Figure 2), the updated framework proved that discovered optimal prices consistently cluster near the true global peak (27.7) across independent seeds, capturing between 90% and 94% of the theoretical maximum profit ($11.925 per customer). Furthermore, a customer allocation and cumulative regret visualization (Figure 4) verified that the incremental caching successfully concentrated the heavy sample budgets on high-yielding prices while restricting exploratory sampling at less optimal pricing.

Figure 4: Customer Allocation & Opportunity Cost per Price Point



When executed on the primary evaluation environment (seed = 2026), the algorithm successfully discovered an optimal price of $28.04, placing it directly adjacent to the true theoretical global peak of $27.70 (Figure 1). Across the entire experiment, the framework tested a total of 34,935 consumers, accumulating $384,525.37 in total revenue, which resulted in an average revenue of $11.007 per customer. Compared to the theoretical maximum profit benchmark of $11.925 per customer, the baseline run captured $92.30 of maximum achievable profit. Across 30 independent simulations with varying

random seeds, the discovered ere prices tightly concentrated around the global peak (Figure 2a), with ECDF distributions confirming profit capture consistently bounded between $90 and $94 (Figure 2b). Perhaps most significantly, the addition of incremental caching for customers constrained testing costs by concentrating over $80 of all customer samples within the near-optimal price zone (25.00–29.00), effectively limiting cumulative regret (Figure 4).